

CS & AI • EXPLAINED

What really happens when you click “Send”?

Your message doesn't travel in one piece. Here's the twist.

THE HOOK

You hit **Send** on a photo. It feels like one whole thing flying across.

But it actually leaves your phone in **thousands of tiny pieces**.

And somehow they all arrive — perfectly. Let's see how. 📱



TODAY'S QUESTION

When you click Send,
how does your message get
there — **in one piece?**

Your message gets chopped up.

"Hi, how are you? 📷" →



- Too big to send as one blob.
- It's split into small chunks called **packets**.
- One photo can become **thousands** of packets.

Each packet carries a **label**.

...

one packet

to: 142.250.195.78

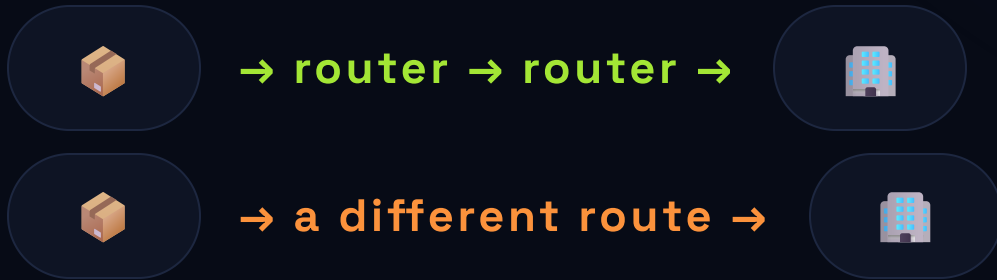
from: your phone

no: 3 of 7

data: ...a chunk of the photo...

- Where it's going + where it's from (**IP addresses**).
- A **sequence number** — "3 of 7".
- And one small piece of the actual data.

Packets travel **separately**.



- Like **dabbawalas**: thousands of tiffins, coded labels, different trains — still reach the right desk.
- Each packet hops router to router, picking its own path.
- So they can arrive **out of order**.

IDEA 4 / 4 — PUT IT BACK TOGETHER

At the end, sorted back in order.



- The **numbers** snap the pieces back in order.
- A packet went missing? It's politely **re-requested**.
- You see the whole message — instantly.

WHY IT MATTERS

This makes the internet **unbreakable**.

- A path goes down? Packets just **reroute** around it.
- Lost a little? Re-send a few packets — that's **buffering**, not total failure.
- One cable carries **millions** of conversations at once — all chopped & mixed.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"My message travels as one whole thing, along one wire, start to finish."

✓ REALITY

It's split into **packets**, each routed independently — maybe by different paths — and reassembled at the end. That system is TCP/IP.

RECAP — 2 THINGS TO REMEMBER

- 1 Your data is split into **packets** — each labelled with addresses + a number.
- 2 They travel **separately** and are reassembled in order at the other end. That's TCP/IP.

Send = chop, address, reassemble. 📁

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Next question: Why do computers only understand 0s and 1s? 📄